

12.3 Latent Diffusion and Text Conditioning

Diffuse in a VAE latent. Condition on a text embedding (often CLIP from Week 5).

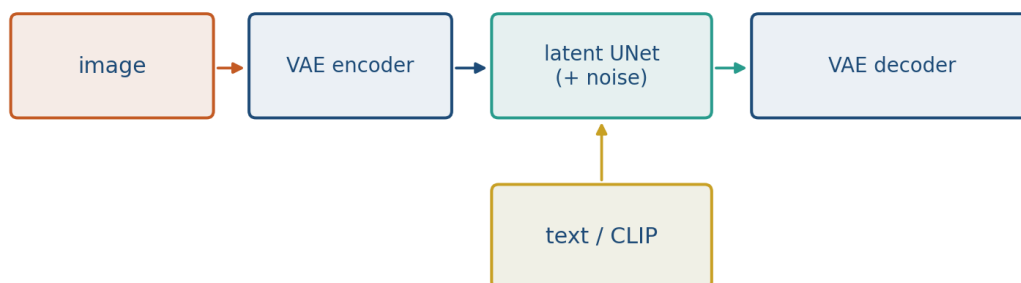
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Pixel-space diffusion on 256×256 RGB is expensive. **Latent diffusion** (Rombach et al.) denoises in a compressed space. **Text** enters as a condition, usually from a frozen CLIP text tower (Week 5).

1. Diffuse in latent space

A pretrained encoder E maps an image to $z_0 = E(x)$. You run the forward process of note **12.1** on z , not on pixels. A decoder D_{VAE} turns the final latent back into an image. Training the U-Net on latents is cheaper; the VAE already threw away some high-frequency detail.

Latent diffusion: denoise a cheap latent, condition on text



Stable Diffusion is this picture plus a big text-conditioned U-Net. The idea is the same at homework scale: compress, denoise, decode.

2. Conditioning with CLIP text

You want $p(x \mid \text{caption})$, not an unconditional generator. Encode the caption with CLIP's **text** tower (note **5.1**). Inject that vector by:

- concatenating it to the U-Net's timestep embedding, or
- **cross-attention**: image (or latent) tokens query the text tokens.

Classifier-free guidance trains with and without the text, then at sample time does

$$\hat{\epsilon} = \epsilon_{\theta}(z_t, t, \emptyset) + s \left(\epsilon_{\theta}(z_t, t, c) - \epsilon_{\theta}(z_t, t, \emptyset) \right).$$

Scale $s > 1$ sharpens adherence to the prompt and can wash out diversity. Report s .

CLIP scores pairs; it does not decode a caption (note **5.2**). Here you only need the text embedding as a handle.

3. What to measure

Text-to-image is not “the picture looks nice.” Report a prompt-alignment score (CLIP cosine of image vs text), a coverage or diversity check, and a failure case (wrong count, wrong object, reading text in the image). Conditioning does not remove Week 5’s bias story: the same CLIP geometry is in the loop.

4. Practice

1. Why denoise latents instead of pixels if you already have a VAE?
2. CLIP’s image tower is optional at sample time for text-to-image. Which CLIP tower is not?
3. Guidance scale $s = 1$ versus $s = 12$: what do you expect to happen to prompt match and to variety?